

TEST-09-FIELD-TRIAL-DESIGN

TEST-09: FIELD TRIAL DESIGN

Carrington Storm Motors — Project AEGIS

Statistical Design • Site Selection • Data Collection • Analysis Plan • Publication Strategy

Document Status: Scientific Field Trial Protocol — IRB-Ready
Version: 1.0 — 15 July 2026
Classification: CONFIDENTIAL — Accredited Investor Eyes Only (Pre-Publication)
Author: Zirconia (Scientific Design) + Specialist Aura Kind (EMBARK — Field Operations)
Statistical Consultant: To be engaged (university partner — target University of Washington Biostatistics)
IRB: To be submitted via partner university

FIELD TRIAL PHILOSOPHY

Laboratory tests prove that a product works under controlled conditions. Field trials prove that it works in the chaos of the real world — with variable installation quality, unpredictable weather, evolving geomagnetic conditions, and the infinite complexity of human behavior. CSM’s field trial program is designed to the standard of a clinical trial in medicine: pre-registered hypotheses, pre-specified analysis plans, independent data monitoring, and publication in peer-reviewed journals regardless of outcome. We publish positive results to prove our products work. We publish negative results to prove our scientific integrity. There are no failed field trials — only field trials that generate data.

PART 1: STATISTICAL DESIGN PRINCIPLES

1.1 Study Design Types by Product

Product	Design	Rationale
Aegis Home	Prospective matched cohort study — 50 Aegis Homes, 25 matched control homes	Cannot randomize (homeowners choose to buy Aegis Home or not); matched controls minimize confounding
Infrastructure (Substations, Bridges)	Before-after study with matched control sites	Cannot build two identical substations — compare each site to its own pre-installation baseline AND to a matched control site that did not receive hardening
Charlemagne Fleet	Single-arm prospective study (1 hardened vessel compared to fleet-wide claims database)	Only 1 vessel hardened — compare its claims/incident experience to the unhardened fleet over same time period
Phantom Robot	Prospective observational study — 10 robots deployed, task completion and reliability measured	Single-arm (no “unhardened robot” exists for comparison — Phantom is the only GIC-hardened robot)
Protonic Network	Single-arm deployment study — 5 nodes, uptime and performance measured	Compare to baseline (conventional radio performance during GIC events from historical data)

1.2 Sample Size Calculations

β (Type II Required Actual

Study	Primary Endpoint	Expected Effect Size	α (Type I Error)	Error / Power)	Sample Size	Sample Size	Margin
Aegis Home — GIC Reduction	Mean GIC current reduction (%) vs. matched control	94% reduction (expected), minimum detectable 80% (conservative)	0.05 (two-sided)	0.20 (80% power)	12 homes per group	50 Aegis + 25 control	4× margin — very well-powered
Aegis Home — Occupant Satisfaction	Proportion “satisfied” or “very satisfied” vs. 80% benchmark	90% (expected)	0.05 (one-sided)	0.20 (80% power)	35 homes	50 homes	Adequate
Infrastructure — GIC Reduction	Mean GIC current reduction (%) vs. pre-installation baseline	90% reduction (expected)	0.05 (two-sided)	0.20 (80% power)	8 sites	10 sites (5 substations + 3 bridges + 2 pipelines)	Adequate
Phantom — Mission Completion Rate	Proportion of missions completed without degradation	95% (expected) vs. 80% benchmark	0.05 (one-sided)	0.20 (80% power)	50 missions	100+ missions (20+ per robot over 6 months)	Adequate
Charlemagne — Insurance Claims Reduction	Rate ratio (claims/vessel-year for hardened vs. fleet baseline)	0.50 (50% reduction)	0.05 (two-sided)	0.20 (80% power)	3 vessel-years (requires multi-year data from 1 vessel or shorter data from multiple vessels)	1 vessel-year from pilot	Underpowered — will report with wide confidence intervals; fleet expansion provides additional vessel-years
Protonic — Uptime	Proportion of time network is operational	99.5% (expected)	0.05 (one-sided)	0.20 (80% power)	8,760 hours (1 year from 1 node)	43,800 node-hours (5 nodes × 8,760 hours)	Well-powered

1.3 Confounding Variable Control

Confounding Variable	Control Method
Geographic GIC Variability	Match control homes/sites within 5 km; same substation feeder; adjust for local geology (conductivity model)
Homeowner Behavior	Standardized instructions; usage logged via Marmalade app; sensitivity analysis excluding non-compliant homes
Seasonal Effects	12-month data collection captures all seasons; analyze by season as secondary endpoint
Installation Quality Variability	Standardized installer training (see ROAD-08 — Installer Training Platform); installation QA checklist (TEST-03 Phase 3)
Measurement Drift	Sensors calibrated pre- and post-study; redundant sensors on critical measurements; data quality flags for suspected sensor failure
Geomagnetic Storm Frequency	Storm occurrence is random — include at least 12 months to capture seasonal and solar cycle variability; supplement with controlled current injection testing (TEST-03 through TEST-07)

PART 2: SITE SELECTION AND PARTNER IDENTIFICATION

2.1 Site Selection Criteria (All Pilots)

Criterion	Weight	Scoring Method
GIC Risk (proximity to geomagnetically active region)	HIGH	K-index frequency analysis; Earth resistivity model; proximity to geomagnetic observatory
Partner Willingness (utility, builder, cruise line)	HIGH	Signed MOU or letter of intent; executive sponsor identified; budget commitment
Regulatory Pathway (permitting feasibility)	HIGH	Pre-application meeting with building department or regulatory body; no known environmental/legal blockers
Data Quality (existing monitoring infrastructure)	MEDIUM	Existing GIC monitoring, weather station, air quality monitoring at or near site
Demographic Representatives	MEDIUM	Match to target market demographics (income, housing type, urban/suburban/rural)
Public Visibility (for marketing/publication)	LOW	Media interest; community engagement potential; educational institution proximity
Logistics (CSM team travel, equipment shipping)	LOW	Within 4-hour travel of CSM EI Segundo office or partner university; commercial shipping feasible

2.2 Partner Identification Process

1. Long-list (20–30 potential partners) from industry databases, conferences, personal networks
2. Screen against criteria → short-list (5–8)
3. Introductory meeting (Steele + Kind) → assess willingness + capability
4. Technical deep-dive (Zirconia + product Agent) → site visit, data review, MOU draft
5. MOU signed → partner officially enrolled in pilot program

Current Partner Identification Status: See ROAD-09 (Partnership Strategy) for detailed pipeline per partner category. Key gap: cruise line partner for Charlemagne pilot is highest priority outstanding partner identification.

PART 3: BASELINE DATA COLLECTION METHODOLOGY

3.1 Pre-Installation Baseline Duration

Study	Baseline Duration	Rationale
Aegis Home	3 months (control homes continue monitoring for full 12 months)	Capture at least one moderate geomagnetic storm ($K_p \geq 5$ occurs ~50 days/year at solar maximum)
Infrastructure	3 months	Same as above; pre-installation data also provides GIC characterization for product design refinement
Charlemagne	12 months (vessel claims/incident data from prior 3 years — retrospective baseline)	Insurance claims are rare events — need multi-year retrospective baseline for statistical power
Phantom	1 month (bench testing at CSM facility)	Phantom is new-build — no field baseline exists; bench testing establishes performance baseline for field comparison

Protonic	None (protonic is novel — no baseline exists)	Compare to conventional radio performance during GIC events from historical utility records
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3.2 Baseline Data Collected

Study	Data	Source	Duration
Aegis Home	GIC currents, energy usage, IAQ, structural strain, homeowner satisfaction (control homes)	Installed sensors + surveys	3 months pre + 12 months concurrent
Infrastructure	GIC currents, transformer neutral current, pipeline CP data, railway signal events	Utility SCADA + installed sensors	3 months pre + 12 months concurrent
Charlemagne	Hull and machinery insurance claims (date, cause, cost), passenger injury claims, voyage delays due to technical issues	Insurance carrier data (anonymized) + cruise line operational data	3 years retrospective
Phantom	Actuator performance, sensor accuracy, navigation accuracy, battery performance, EMI susceptibility	CSM bench test data	1 month (bench) + 6 months (field)
Protonic	Weather data, geomagnetic activity, conventional radio uptime in test regions (from utility records)	NOAA SWPC + utility records	12 months concurrent

PART 4: INTERVENTION DEPLOYMENT PROTOCOL

4.1 Installation Quality Control

All pilot installations follow the QA protocols defined in TEST-03 (Phase 3) for Aegis Home, TEST-04 for Charlemagne, TEST-06 for infrastructure. Field trial data is segregated between correctly-installed products and those with installation defects (identified during QA) for sensitivity analysis.

4.2 Blinding

- **Single-blind:** Homeowners in control homes do not know their home’s GIC measurements (no real-time display)
- **Open-label:** Aegis Home owners know their home is hardened (impossible to blind — the products are physically visible)
- **Blinded analysis:** Statistician analyzing data is blinded to home assignment (Aegis vs. control) until analysis is complete

PART 5: MONITORING AND DATA COLLECTION PLAN

5.1 Data Collection Infrastructure

Component	Technology	Notes
Sensors	Per TEST-03 (Aegis Home), TEST-04 (Charlemagne), TEST-06 (Infrastructure)	All NIST-traceable calibration
Data Aggregation	On-site NI CompactDAQ or Raspberry Pi with ADC hat → local	Store-and-forward architecture tolerates internet outages

Cloud Platform	NAS → cloud sync (AWS S3) AWS S3 → PostgreSQL + PostGIS → CSM Digital Twin API	Encrypted at rest (AES-256) and in transit (TLS 1.3)
Monitoring Dashboard	LandOLil v10+ with real-time streaming for CSM engineers; Grafana for detailed technical dashboards	Alert thresholds set for sensor failure, data gap >1 hour, GIC event detection
Mobile App	Marmalade (homeowners), LandOLil (utility/installer partners)	Offline-first architecture; sync when internet available

5.2 Data Quality Assurance

Check	Frequency	Action if Failed
Sensor Calibration Verification	Pre-installation, post-installation, quarterly during study	Recalibrate sensor; flag data between last good calibration and recalibration
Data Completeness	Daily automated check (expect >95% completeness)	Investigate missing data; dispatch technician if hardware failure
Sensor Drift Detection	Weekly: compare redundant sensor pairs; flag if divergence >3σ of expected variation	Investigate; recalibrate or replace sensor
Outlier Detection	Automated: z-score >5 triggers alert	Manual review; if confirmed physical event (actual GIC event), retain; if sensor artifact, flag/exclude
Timestamp Accuracy	NTP sync all data loggers; verify hourly	Correct clock drift; flag data if timestamp uncertainty >1 second

5.3 Adverse Event Reporting

Event Type	Reporting Timeline	Action
Serious Adverse Event (SAE) — Electrical shock, fire, structural failure, injury, death	Within 24 hours to Zirconia + Steele + IRB (if human subjects involved)	Immediate investigation; root cause analysis; product recall/field modification if indicated
Product Malfunction — Any CSM product failing to meet specification	Within 72 hours to product Agent + Chester	Diagnostic investigation; repair/replace; trend analysis
Data Loss — Sensor failure, data gap >24 hours	Within 48 hours to Zirconia	Restore from backup; repair sensor; document gap in final report
Protocol Deviation — Any departure from field trial protocol	Within 1 week to Zirconia + IRB	Document deviation; assess impact on data validity

PART 6: STATISTICAL ANALYSIS PLAN

6.1 Primary Analysis (Intent-to-Treat)

All enrolled homes/sites are included in analysis regardless of protocol deviations, product malfunctions, or homeowner withdrawal. This is the “real world” estimate — it includes the effect of installation quality variability, homeowner behavior, and product reliability. Pre-specified before any data is collected.

6.2 Secondary Analysis (Per-Protocol)

Excluding homes/sites with protocol deviations (installation defects identified during QA, homeowner non-compliance with sensor maintenance, early withdrawal). This estimates product efficacy under “ideal” conditions.

6.3 Subgroup Analyses (Exploratory)

Subgroup	Hypothesis	Pre-Specified?
Geographic Region	GIC reduction varies by region (geology, grid configuration)	Yes

Housing Type	GIC reduction differs between single-family and multifamily	Yes
Season	GIC reduction varies by season (frozen ground affects GIC)	Yes
Homeowner Income	Satisfaction, health outcomes differ by income (equity analysis)	Yes
Geomagnetic Storm Intensity	GIC reduction is consistent across storm magnitudes (Kp 5–9)	Yes
Installation Quality (QA Score)	Higher QA score → better GIC reduction	Yes (exploratory)

All subgroup analyses reported with interaction p-values and 95% confidence intervals. No post-hoc “p-hacking” — exploratory analyses clearly labeled as such.

6.4 Missing Data Handling

Missing Data Mechanism	Handling Method
Missing Completely at Random (MCAR) — sensor randomly fails	Complete case analysis (primary); multiple imputation (sensitivity analysis)
Missing at Random (MAR) — sensor fails more often in winter (battery drain)	Multiple imputation using weather, season, and operational variables as predictors
Missing Not at Random (MNAR) — homeowner withdraws because they’re dissatisfied	Worst-case imputation (assume dissatisfied homeowners would have had poor outcomes); report both optimistic and conservative estimates

PART 7: REPORTING AND PUBLICATION PLAN

7.1 Pre-Registration

All field trials pre-registered on Open Science Framework (osf.io) or ClinicalTrials.gov (for trials with human health outcomes — e.g., Aegis Home health monitoring, Cervical-Guard driver study) before data collection begins. Pre-registration includes: hypotheses, primary/secondary endpoints, sample size justification, analysis plan, and subgroup analyses.

7.2 Publication Timeline

Study	Data Collection Complete	Analysis Complete	Manuscript Submitted	Target Journal
Aegis Home Pilot	Q4 2028	Q1 2029	Q2 2029	Building and Environment or IEEE Trans. EMC
Infrastructure Pilot	Q4 2028	Q2 2029	Q3 2029	IEEE Trans. Power Delivery
Charlemagne Pilot	Q2 2030	Q3 2030	Q4 2030	Marine Structures or Ocean Engineering
Phantom Pilot	Q2 2029	Q3 2029	Q4 2029	IEEE Robotics and Automation Letters
Protonic Pilot	Q4 2028	Q1 2029	Q2 2029	Nature Communications or IEEE Comm. Mag

7.3 Data Sharing

After primary publication, de-identified data deposited in: - **Dryad** (general scientific data repository) - **CSM Digital Twin public API** (real-time GIC monitoring data for researchers) - **NOAA SWPC** (GIC event data contributed to space weather database) - Embargo period: Data available to journal reviewers during peer review; public release upon publication

FIELD TRIAL BUDGET SUMMARY (from ROAD-06)

Study	Total Budget	Notes
Aegis Home (50 homes)	\$6.2M	Largest study; includes home construction cost
Infrastructure (10 sites)	\$8.5M	Includes partner co-funding from utilities/DOTs
Charlemagne (1 vessel)	\$50.8M	Vessel retrofit (\$47M) is capex; trial monitoring (\$3.8M) is opex
Phantom (10 robots)	\$8.5M	Includes robot build cost
Protonic (5 nodes)	\$3.8M	Lowest-cost pilot
Aegis Embark (100 vehicles)	\$2.8M	Consumer product pilot — largest N, lowest per-unit cost
TOTAL	\$80.6M	

“A field trial is a promise of transparency. We will tell the world what we are going to measure before we measure it. We will publish the results regardless of what they show. We will share the data with anyone who wants to verify our analysis. This commitment costs us nothing in competitive advantage — because our competitive advantage is not in keeping secrets, it is in making products that work. If our products work, the data will show it. If they do not, we deserve to know, and our customers deserve to know, before they buy.” — Zirconia, Director of Test & Validation

Document generated 2026-07-15. All statistical analysis plans are preliminary and will be finalized with independent biostatistician review before data collection begins. IRB applications to be submitted through partner university (University of Washington target) for all studies involving human subjects data collection (homeowner surveys, health outcome monitoring, Phantom human-robot interaction). Pre-registration on Open Science Framework to be completed before first baseline data is collected for each study.